Hybrid EEG-EMG Brain-Machine Interface

A Project Roadmap: Signal Selection, Processing, Machine Learning, and Edge Deployment on STM32

Prepared for: Amogh | Final-year engineering capstone
Hardware: OpenBCI Cyton (8-ch, ADS1299) | 3D-printed comb-electrode headset | STM32 F446ZET6 (Cortex-M4F)
Target modalities: 8-channel EEG + 4-channel EMG | Goal: hybrid task control
Timeline: August - November 2026 | Date: 3 August 2026

What this document is. A build plan you can follow week by week, plus the research behind each choice. It tells you which signals to record, how to clean and turn them into features, which classifiers and fusion methods to use, how to shrink the model onto the STM32, and how to shape the work into one or two papers. Every technical claim is backed by a numbered reference at the end.

Reality check up front (read this first). You want 8 EEG + 4 EMG = 12 channels, but a single 8-channel Cyton gives you 8. This channel budget is the first decision in the plan, and it drives your sample-rate and paradigm choices. Section 3 lays out the three ways to solve it. Treat the rest of the roadmap as conditional on that call.

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1. Executive summary

The aim is a wearable system that reads brain and muscle signals, decides what the user wants, and drives an application in real time. EEG (from the scalp) carries slow, noisy signals about intent and visual attention; EMG (from muscle) carries fast, strong signals about movement. Used together they cover each other's weak spots, and the published record shows the pair beats either one alone. One hand-exoskeleton study reports 90% accuracy for an EEG-EMG scheme against about 80% for EEG-only spatial patterns; a wheelchair study using an LSTM on the fused signal reports 97% [5,7].

Recommended target. Build a hybrid controller that sends a small, fixed set of commands - forward, left, right, stop - to a **simulated** wheelchair or drone first, then to real hardware once the decoder is stable. Simulation removes the robotics risk while you get the signals working, which is where the science and the papers live. EMG gestures give you fast, reliable discrete commands; EEG (steady-state visual evoked potential, SSVEP) gives you a hands-free selection channel; a muscle 'clench' works as a safety stop. This split is exactly how the hybrid-UAV literature builds robust control [11,15].

A prosthetic or robotic hand is the same pipeline with the same code and is arguably the more in-demand application; the roadmap keeps that door open because switching target is a change of output mapping, not of signal processing.

The three paper angles you picked - fusion accuracy, edge/TinyML on the STM32, and a signal-processing and classifier comparison - are not three separate projects. They are three slices of one build, and Section 12 shows how the same experiments feed all three. Everything is scoped to be finished by a final-year student between August and November.

2. System overview and signal chain

The system is a straight pipeline. Signals flow left to right; only the fusion and control blocks need the two streams at once.

Stage	EEG branch	EMG branch
1. Sense	8 scalp electrodes (comb, dry/gel)	4 electrodes on forearm / temple muscles
2. Amplify + digitise	ADS1299, 24-bit, on the Cyton	Same board, same converter
3. Stream	BrainFlow / Lab Streaming Layer to PC	Same link, time-aligned
4. Clean	Band-pass, notch, re-reference, de-artifact	Band-pass 20-450 Hz, rectify/envelope
5. Window	0.5-2 s epochs, overlapping	150-250 ms sliding windows
6. Features	Band power / CSP / CCA / covariance	Hudgins time-domain set (MAV,WL,ZC,SSC)
7. Classify	SSVEP freq / motor-imagery class	Gesture class (rest/grasp/left/right)
8. Fuse + decide	Decision-level fusion -> single command	(shared with EEG)
9. Act	Command -> simulator, then robot/drone	Clench -> hardware safety stop

Stages 1-3 are the same board for both branches, which is the whole appeal of the Cyton: one 24-bit ADS1299 front end records EEG and EMG at once. The branches split at cleaning (stage 4) because the two signals live in different frequency bands, and rejoin at fusion (stage 8).

3. Hardware and the channel-budget decision

OpenBCI Cyton. Eight channels built on the TI ADS1299, 24-bit resolution per channel, default sample rate 250 Hz, streamed over the RFDuino radio dongle. The same input can be wired single-ended (referential) or bipolar, and the board is rated for EEG, EMG and ECG - so one board covers both your modalities [19]. The ADS1299 itself can run far faster, but the radio link caps you at 250 Hz for 8 channels; the WiFi shield raises that ceiling to about 1000 Hz for 8 channels [19].

The problem. You want 8 EEG + 4 EMG = 12 channels. One Cyton has 8. You must resolve this before you print electrodes or write acquisition code, because it fixes both your channel count and your sample rate, and those two fight each other for EMG.

Option	Channels	Sample rate	EMG quality	EEG quality	Verdict for this project
A. Cyton only, split 4 EEG + 4 EMG	8	250 Hz (1 kHz with WiFi shield)	Good - full 20-450 Hz band if >=1 kHz	Fewer sites; enough for SSVEP at Oz/O1/O2	Simplest. Best if EMG-led control and SSVEP.
B. Cyton + Daisy (16-ch)	16 (use 8 EEG + 4 EMG)	125 Hz over dongle	Weak - 125 Hz only reaches ~62 Hz, loses EMG spectrum	Full 8-ch EEG montage, 125 Hz is fine	Gets your stated 8+4 but starves EMG.
C. Cyton + Daisy + WiFi shield	16	up to ~250-500 Hz	Marginal-OK for envelope features	Full montage	Best of both, extra cost and setup.

The tension in one line. EEG needs many channels but little bandwidth (mu/beta rhythms stop near 30 Hz, SSVEP harmonics near 40 Hz), so 125 Hz sampling is fine. EMG needs bandwidth (its energy runs 20-450 Hz), so it wants >=1000 Hz sampling but only a few channels. Adding the Daisy buys channels at the cost of the sample rate EMG needs.

Recommendation. Start with **Option A** (Cyton only, 4 EEG + 4 EMG, WiFi shield at ~1 kHz). It keeps EMG clean, gives you enough occipital channels for SSVEP and enough central channels for a first motor-imagery test, and it is the lightest thing to get streaming. Move to Option C only if a later experiment truly needs the full 8-channel EEG montage. Note this decision in the paper - a channel/sample-rate trade study is itself publishable content for the comparison paper.

STM32 F446ZET6. ARM Cortex-M4F at 180 MHz with a hardware floating-point unit, 512 KB Flash, 128 KB SRAM. That is comfortably enough to run the classifiers in this plan (Section 9). The FPU matters: your feature maths (covariance, filtering, dot products) runs in float without penalty, and ARM's CMSIS-DSP/CMSIS-NN libraries are tuned for exactly this core [14,17].

4. Which signals are most suitable

This is the heart of your question. There are three mainstream EEG control paradigms and one EMG approach. They differ in how much they trust the user's brain versus a screen, how much training they need, and how fast and accurate they are. Pick by matching them to your control target.

4.1 EEG paradigms

Paradigm	What the brain does	Where to record	Speed / accuracy	Training burden	Fit here
SSVEP (steady-state visual evoked potential)	Stares at a target flickering at a set frequency; the visual cortex 'rings' at that frequency	Occipital: Oz, O1, O2	High. 75-100% accuracy, ITR 12-150 bit/min reported [9,10,3]	Almost none - works nearly out of the box	Best first EEG channel. Reliable, fast, little training.

Paradigm	What the brain does	Where to record	Speed / accuracy	Training burden	Fit here
P300	Reacts to a rare, expected stimulus with a positive bump ~300 ms later	Parietal: Pz, Cz	Good accuracy, slower (needs repeats)	Low-moderate	Good for menu/speller, slower than SSVEP.
Motor imagery (ERD/ERS)	Imagines moving a hand/foot; mu(8-12)/beta(13-30) power drops then rebounds	Sensorimotor: C3, Cz, C4	Moderate. ~67-75% for 4-class is a strong baseline [2,4]	High - days of user training	Research-rich but hardest to hit by November.

Reading this table for your project. SSVEP is the workhorse: it is fast, accurate, needs no user training, and its detection method (canonical correlation analysis, below) is simple enough to run on the STM32. P300 is a fine second option for a menu. Motor imagery is the most 'brain-computer interface' of the three and the richest for a paper, but it is the least reliable and needs the most training - treat it as a stretch goal or a comparison arm, not your main control channel.

4.2 EMG - the muscle channel

Surface EMG is the opposite of EEG: large amplitude, high signal-to-noise, fast, and easy to classify. For a hybrid controller it carries the discrete commands. Two ways to use it:

- **Gesture recognition.** Place electrodes over forearm flexor/extensor muscles and classify hand postures (rest, fist, wrist-left, wrist-right, pinch). With the classic time-domain feature set this reaches ~90% in under a second [12]. This is your main command source.
- **Activation / clench as a switch.** A jaw or hand clench is a clean, unmistakable burst. Use it as a 'select', 'enable', or emergency stop - the hybrid-UAV papers use exactly a teeth-clench as an interrupt [11].

4.3 The recommended hybrid scheme

Channel	Signal	Job in the controller
Primary command	EMG gesture (4-class)	forward / left / right / rest - the moment-to-moment control
Selection / mode	EEG SSVEP (2-4 flicker freqs)	hands-free menu or mode switch (e.g. speed, target)
Safety / confirm	EMG clench burst	'stop' and 'confirm' - always available, near-zero error
(stretch) Intent gate	EEG motor imagery or attention	enable/disable the whole controller from brain state

This mirrors how the strongest hybrid systems are built: let each signal do what it is good at, and fuse at the decision stage so a weak read on one channel does not sink the command [5,11]. It also gives you a clean ablation for the fusion paper - EMG-only, EEG-only, and fused - on the same task.

5. Acquisition and preprocessing

Garbage in, garbage out applies harder to EEG than almost anywhere. A disciplined cleaning pipeline is what separates a demo that works from one that does not. Do the same core steps every time and log them, so results are reproducible for the paper [20].

5.1 EEG cleaning

- **Band-pass filter.** Keep 1-40 Hz for most work; this holds mu, beta and low SSVEP while dropping drift and high-frequency muscle noise [20].
- **Notch filter.** Remove mains hum at 50 Hz (India) with a narrow notch - the standard first defence against power-line pickup [20].
- **Re-reference.** Common Average Reference (CAR) - subtract the mean of all channels - cuts shared noise and sharpens local activity [20].

- **Artifact handling.** Independent Component Analysis (ICA) separates and removes eye-blink and muscle components; for a real-time loop, use a lighter automatic rejection or a trained ICA unmixing matrix applied online [20].

5.2 EMG cleaning

- **Band-pass 20-450 Hz** - the useful surface-EMG band; this is why Option A's high sample rate matters.
- **Notch at 50 Hz** for mains, same as EEG.
- **Rectify and envelope** (full-wave rectify then low-pass, or take a moving RMS) when you want an activation level rather than the raw waveform - handy for the clench detector.

5.3 Windowing

Cut the continuous stream into short, overlapping windows and classify each. For EMG, 150-250 ms windows keep the control feel responsive (the Hudgins set is deliberately insensitive to window length [12]). For EEG SSVEP, 1-2 s windows trade speed for a cleaner frequency estimate; longer windows raise accuracy but slow the interface, so tune this and report the curve.

6. Feature extraction (EEG and EMG)

Features turn a window of raw samples into a short vector a classifier can use. Match the feature to the paradigm.

6.1 EEG features

Method	Paradigm	Idea	Note
Band power	MI, general	Power in mu/beta bands per channel	Simple, cheap, weak spatial info
CSP / FBCSP	Motor imagery	Spatial filters that maximise class variance difference; FBCSP does it per frequency sub-band	FBCSP won BCI Competition IV (2a ~67%) [2]
CCA / FBCCA	SSVEP	Correlate signal with sine/cosine templates at each flicker frequency; pick the best	FBCCA adds harmonics, big ITR gain [3]
Riemannian covariance	MI, P300, general	Use the channel covariance matrix as the feature on its curved (SPD) space	Robust to noise, minimal tuning, contest-winning [4]

Guidance. For SSVEP use CCA, then FBCCA once it works - it is nearly parameter-free and runs on the MCU. For motor imagery, start with CSP+band-power (the textbook baseline) and add a Riemannian classifier as your 'strong' arm; Riemannian methods are the current state of the art for robustness and need little tuning [4].

6.2 EMG features

Use the **Hudgins time-domain set**: Mean Absolute Value (MAV), Waveform Length (WL), Zero Crossings (ZC), and Slope Sign Changes (SSC). Four numbers per channel, cheap to compute, and they capture amplitude, frequency and complexity at once. This set is popular because it gives high accuracy, is insensitive to window size, and is light enough for a microcontroller [12]. Add Root Mean Square (RMS) and a short auto-regressive model if you want a few more features; studies report ~90% gesture accuracy in ~800 ms with these [12].

7. Classification and sensor fusion

7.1 Classifiers

Classifier	Strength	Cost on MCU	Use for
LDA (linear discriminant)	Fast, stable, tiny, the BCI default	Trivial	First baseline for MI and EMG
SVM	Strong on small data, good margins	Small (linear) / moderate (RBF)	EMG gestures, SSVEP colour studies [9]
Random forest	Handles mixed features, robust	Moderate	SSVEP feature sets, EMG
Riemannian MDM	Robust, deterministic, little tuning	Small-moderate	MI / P300 'strong' arm [4]
EEGNet (compact CNN)	Learns features, one net across paradigms, ~2 orders fewer params than big CNNs	Fits M4 after int8 quantisation [1,16]	End-to-end EEG, the deep-learning arm

Guidance. Build the ladder in this order: LDA first (get the loop closed), then SVM/Random Forest, then the two 'strong' arms - Riemannian for classical robustness and EEGNet for the deep-learning story. EEGNet is deliberately tiny and has already been put on Cortex-M4/M7 parts with ST's tools, so it is a realistic on-chip target, not just a PC model [1,16].

7.2 Fusion - how to combine EEG and EMG

Level	What is combined	Pros	Cons
Data-level	Raw/filtered samples stacked	Keeps all information	Sensitive to noise, needs tight time-align
Feature-level	Feature vectors concatenated	Simple, one classifier, strong accuracy	One bad channel can drag the vector
Decision-level	Each modality's class + confidence	Robust; a weak channel is out-voted; modular	Needs a fusion rule (weighted vote, Bayesian)

Recommendation: decision-level fusion. Classify EEG and EMG separately, then combine the two decisions with a confidence-weighted rule. It is the most robust, the easiest to reason about, and it lets you drop in a fatigue- or reliability-weighted vote - a trick recent hybrid papers use to stay accurate over long sessions [5,8]. It also gives you the cleanest ablation table for the fusion paper: EEG-only vs EMG-only vs fused, same task, same subjects.

7.3 How to score it

- **Accuracy** and **Cohen's kappa** - kappa corrects for chance and is standard in BCI reporting.
- **Information Transfer Rate (ITR, bits/min)** - the field's speed-and-accuracy number; report it so your results compare to the literature [3,10].
- **Confusion matrices** per class, and a **window-length vs accuracy** curve - both are expected in a paper.
- **Cross-validation done right** - never let windows from one trial land in both train and test; use trial-wise or session-wise splits or you will report inflated numbers.

8. Public datasets to build and benchmark on

Do not wait for your own headset to start the software. Build and validate the whole pipeline on public data first, then swap in your recordings. This de-risks the schedule and gives the comparison paper a standard benchmark reviewers recognise.

Dataset	Signal	Contents	Use it for
BCI Competition IV-2a	EEG	9 subjects, 4-class motor imagery (L/R hand, feet, tongue), 22 ch, 250 Hz [2]	MI pipeline, CSP/FBCSP/Riemannian/EEGNet baselines

Dataset	Signal	Contents	Use it for
BCI Competition IV-2b	EEG	Left/right hand MI, 3 bipolar channels	2-class MI, closest to your channel count
Ninapro DB5	EMG	40 gestures, 40 subjects, 8-ch forearm sEMG at 2 kHz	EMG gesture pipeline and feature/classifier study
PhysioNet EEG MI	EEG	109 subjects, motor movement/imagery, 64 ch	Large-scale pre-training for EEGNet

Ninapro DB5's 8-channel forearm montage is close to what you will actually wear, which makes it the right sandbox for the EMG side [13]. For EEG, IV-2a is the reference every MI paper cites, and IV-2b's low channel count is the closest match to your hardware.

9. Edge deployment on the STM32 F446ZET6

This is your TinyML paper. The question is not 'can a model run on the chip' - it is 'how small, how fast, and how little power', measured properly. The tooling is mature.

- **X-CUBE-AI** - ST's free expansion pack. It takes a trained Keras/TFLite model, optimises it, and generates C code for the STM32; it reports the Flash, RAM and cycle cost before you flash [14,15].
- **CMSIS-NN / CMSIS-DSP** - ARM's hand-tuned kernels for Cortex-M. Int8 inference through CMSIS-NN is markedly faster than plain C (one study: 7.4 ms vs 14.2 ms for the same net) [17].
- **Quantisation** - convert float weights to 8-bit integers after training. This shrinks the model ~4x and speeds it up with little accuracy loss; it is the standard route to fit a CNN on an MCU [18].

9.1 What fits

Model	Rough size	Fits F446 (512 KB Flash / 128 KB SRAM)?	Note
LDA / linear SVM	< 5 KB	Yes, trivially	Just a matrix multiply; start here
CSP + LDA	< 20 KB	Yes	Spatial filter + linear step, very cheap
Riemannian MDM	10s of KB	Yes	Covariance + distance to class means
EEGNet (int8)	~ tens of KB	Yes	Already demonstrated on M4/M7 with X-CUBE-AI [1,16]
Hudgins + RF (EMG)	10s-100s KB	Yes (prune trees)	Watch node count; cap depth

What to measure for the paper. For each model: on-chip **latency** per window (from CMSIS cycle counter), **Flash and SRAM** use (X-CUBE-AI report), **accuracy** vs the PC version after quantisation, and, if you can, **power/energy** per inference. Plot the accuracy-vs-cost trade-off across the model ladder - that single figure is the spine of the TinyML paper [16,17].

9.2 Realistic split of work

The PC handles training and heavy exploration. The STM32 runs the final, frozen decoder in the real-time loop: read a window over UART/SPI from the Cyton (or from a logged file for the benchmark), compute features, run the quantised classifier, emit a command. Keep the fusion rule on-chip too - it is only a weighted vote. Start by porting the cheapest model (LDA), prove the loop end-to-end, then climb the ladder to EEGNet.

10. Project roadmap: August to November

About fourteen weeks. The plan front-loads software on public data so hardware delays cannot stall you, and it reaches a working single-modality demo before attempting fusion. Each phase ends in something you can show or measure.

Phase / weeks	Dates	Goals	Deliverable
0. Setup (wk 1)	Aug 4-10	Assemble Cyton + headset, decide channel budget (Sec 3), install BrainFlow/LSL + Python stack, first recording	Live EEG+EMG stream, impedance check
1. Signal validation (wk 2-3)	Aug 11-24	Confirm real signals: alpha blocking (eyes open/close), an SSVEP response, EMG bursts on gesture; log SNR	Signal quality report + clean recordings
2. Offline pipeline (wk 4-6)	Aug 25-Sep 14	Build preprocess->feature->classify on IV-2a/2b + Ninapro DB5; get LDA/CSP/SSVEP-CCA/EEGNet baselines	Reproducible pipeline + baseline table
3. Own-data + single-modality online (wk 7-8)	Sep 15-28	Record your own protocol; run live SSVEP (FBCCA) and live EMG (Hudgins+LDA) decoders	Two working real-time decoders
4. Hybrid fusion + control loop (wk 9-11)	Sep 29-Oct 19	Decision-level fusion; drive a wheelchair/drone simulator; run EEG-only/EMG-only/fused ablation	Hybrid demo + fusion results (Paper 1 data)
5. STM32 port + benchmark (wk 12-13)	Oct 20-Nov 2	Quantise, deploy via X-CUBE-AI, measure latency/memory/power across the model ladder	On-chip decoder + benchmark table (Paper 2 data)
6. Write-up + demo (wk 14+)	Nov 3-30	Comparison study, real-hardware stretch, polish; draft the papers and final report	Papers 1-2 drafts, final demo, report

Sequencing rule. Do not touch fusion (Phase 4) until both single-modality decoders work (Phase 3), and do not touch the STM32 (Phase 5) until the PC decoder is frozen. Chasing all three at once is the classic way these projects miss the deadline.

11. Risk register

Risk	Likelihood	Impact	Mitigation
Channel budget starves EMG (Daisy at 125 Hz)	High	High	Prefer Option A (Cyton, 1 kHz WiFi shield); use envelope features if forced to 125 Hz
Dry comb electrodes give poor EEG contact	Medium	High	Test impedance early; add a little gel/saline; keep occipital sites for SSVEP where SNR is highest
Motor imagery accuracy too low	Medium	Medium	Lean on SSVEP + EMG for control; keep MI as a comparison arm, not the main channel
Mains/EMG artifact swamps EEG	Medium	Medium	Notch + CAR + ICA; record away from power supplies; twist leads
EEGNet won't fit or is slow on MCU	Low	Medium	Int8 quantise; fall back to CSP+LDA or Riemannian - all fit easily
Real robot/drone integration slips	Medium	Low	Simulator is the primary target; hardware is a stretch goal
Schedule slip from doing all 3 papers at once	Medium	High	Follow the phase gates; one build feeds all three angles

12. Turning this into 1-2 papers

You picked three angles. They come from **one** set of experiments, so aim for two solid papers (a fusion paper and a TinyML paper) with the comparison study either folded into them or spun out as a short third. Here is how the same data serves all three.

Angle	Core claim	Experiments (already in the roadmap)	Where to send it
P1. EEG-EMG fusion accuracy	Decision-level fusion beats EEG-only and EMG-only on a real control task	Phase 4 ablation: EEG-only / EMG-only / fused, with accuracy, kappa, ITR	Student/UG conference; IEEE EMBC or a workshop; MDPI Sensors/Biosensors
P2. TinyML edge BCI on STM32	A hybrid decoder runs in real time on a Cortex-M4 within stated latency/memory/power	Phase 5 benchmark across the model ladder; accuracy-vs-cost figure	Embedded/TinyML venue; IEEE sensors/embedded workshops
P3. SP + classifier comparison	Systematic comparison of features (CSP/FBCSP/CCA /Riemannian, Hudgins) and classifiers for this low-cost rig	Phase 2-3 baselines on public + own data; channel/sample-rate trade study (Sec 3)	Fold into P1/P2, or short paper / poster

12.1 What makes it publishable at your level

- **A clean ablation.** The EEG-only / EMG-only / fused comparison on one task is a complete, honest result - reviewers value that more than a fragile record number.
- **Real measurements on real hardware.** On-chip latency/memory/power numbers are concrete and hard to argue with; few student projects report them, which is your edge.
- **Reproducibility.** Public datasets + released code + a low-cost open rig (OpenBCI + a 3D-printed headset) is itself a contribution; say so, and release the design files.
- **Honest limitations.** Small subject count, dry-electrode noise, the 125 Hz EMG trade-off - naming these well is what a 4th-year paper is expected to do.

12.2 Novelty handles worth exploring

- Fatigue- or confidence-weighted decision fusion that adapts over a long session [5,8].
- Channel/sample-rate co-design: how few channels and how low a rate can you use and still control the task - directly useful for cheap edge hardware.
- Quantisation-vs-accuracy for BCI models specifically (most TinyML papers use vision/audio, not EEG/EMG) [16,18].

13. Bill of materials and setup checklist

13.1 You have / assume you have

- OpenBCI Cyton (8-ch, ADS1299) + RFDuino USB dongle.
- 10 comb electrodes + the 3D-printed headset (OpenBCI Ultracortex-style frame).
- STM32 F446ZET6 (Nucleo-144 or custom board).
- EMG electrodes (gel/Ag-AgCl or dry) + leads for forearm and temple/jaw.

13.2 Add / confirm

- **OpenBCI WiFi shield** - to lift the sample rate to ~1 kHz for clean EMG (Option A). Strongly recommended.
- **Daisy module** - only if you commit to the full 8-channel EEG montage (Option B/C).
- SSVEP stimulus: a monitor or a set of LEDs driven at fixed frequencies (e.g. 8, 10, 12, 15 Hz).
- Electrode gel/saline, measuring tape and skin-prep for good contact.
- A wheelchair/drone **simulator** (a simple 2-D sim or a game engine target) before any real platform.

13.3 Software stack

- **BrainFlow** or **Lab Streaming Layer (LSL)** to pull data off the Cyton with timestamps.
- **Python**: NumPy/SciPy, MNE-Python (EEG), scikit-learn, **pyRiemann** (Riemannian), TensorFlow/Keras (EEGNet).
- **STM32CubeIDE** + **X-CUBE-AI** + **CMSIS-NN/DSP** for the embedded side.

14. References

All links verified from literature searches on 3 August 2026. Where a paper has both a journal and a preprint, the more accessible link is given.

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